PF Lab #3 Assignment

Name = Muhammad Bilal

Roll No =26P-3089

Secation = BS SE1B

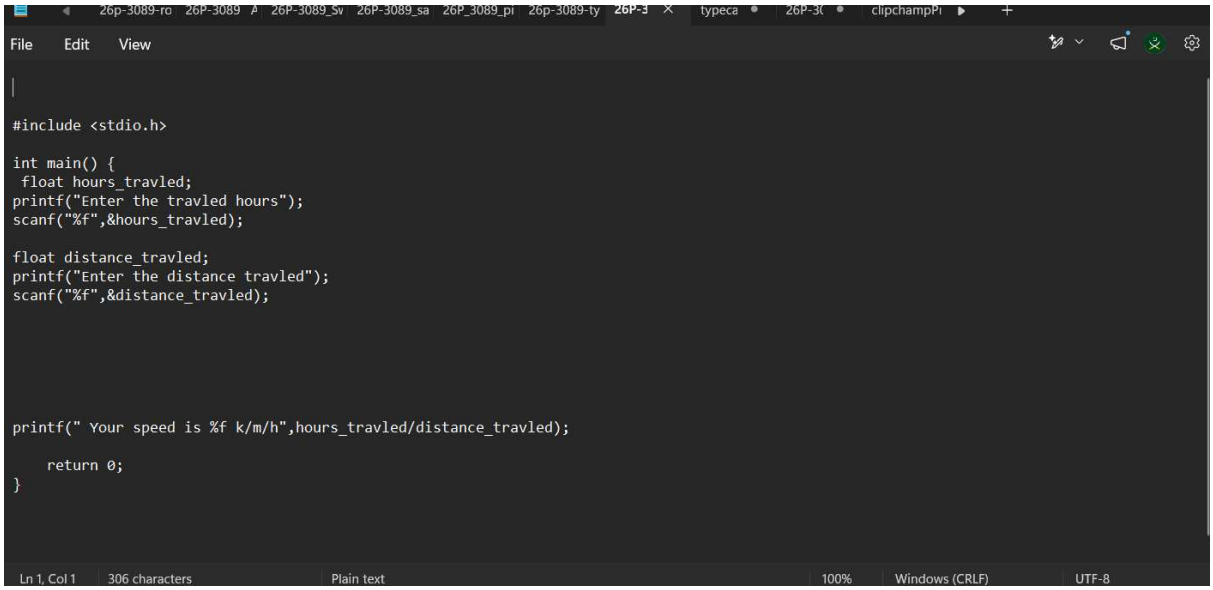
QUESTION#1: Average Speed

Code :-

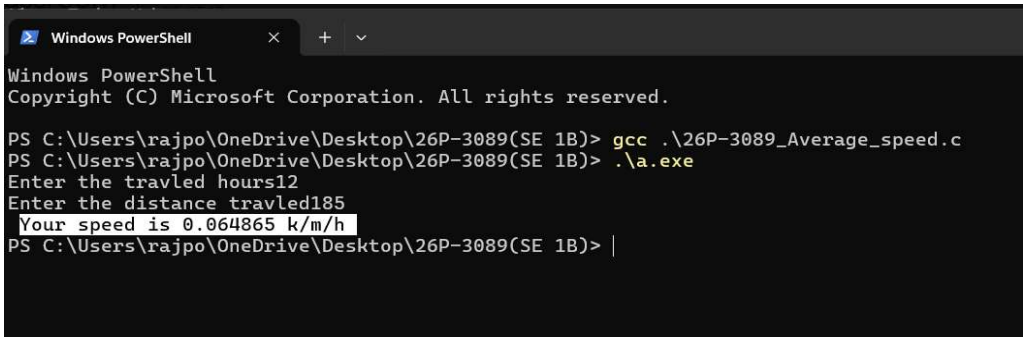
```
#include <stdio.h>
int main() {
float hours_travled;
printf("Enter the travled hours");
scanf("%f",&hours_travled);

float distance_travled;
printf("Enter the distance travled");
scanf("%f",&distance_travled);
printf(" Your speed is %f k/m/h",hours_travled/distance_travled);
return 0;
}
```

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OUTPUT ON TERMINAL



QUESTION#2 Swapping Two Values

Code :-

```
# include <stdio.h>

int main() {

int num1, num2, num3;

 printf("Enter num1:");

 scanf("%d", &num1);

 printf("Enter num2:");
```

```
scanf("%d", &num2);


num3= num1;

num1= num2;

num2=num3;

printf("After swapping:\n");

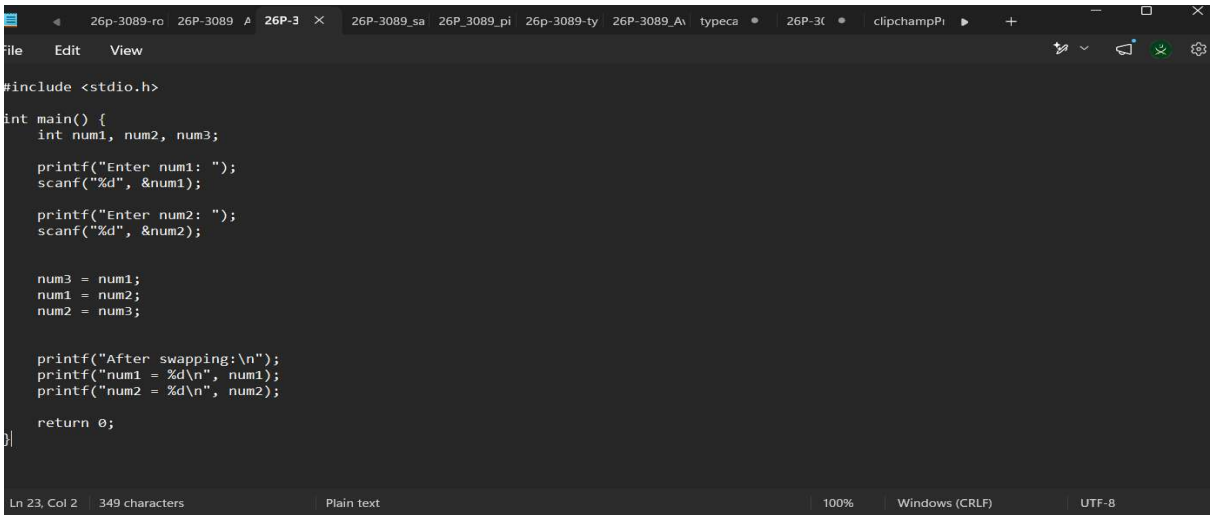
 printf("num1 =%d\n", num1);

printf("num2=%d\n", num2);

Return 0;

}
```

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OUTPUT ON TERMINAL

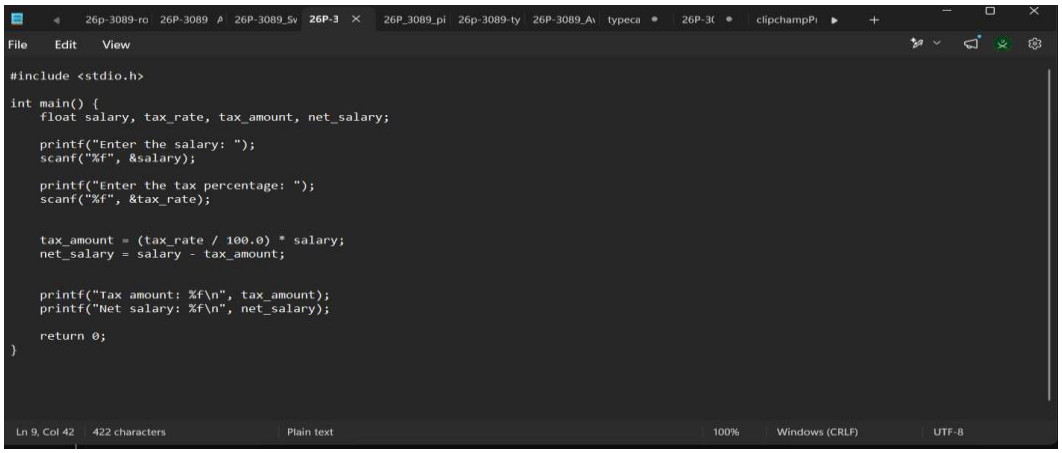


QUESTION#3 SALERY-TEX

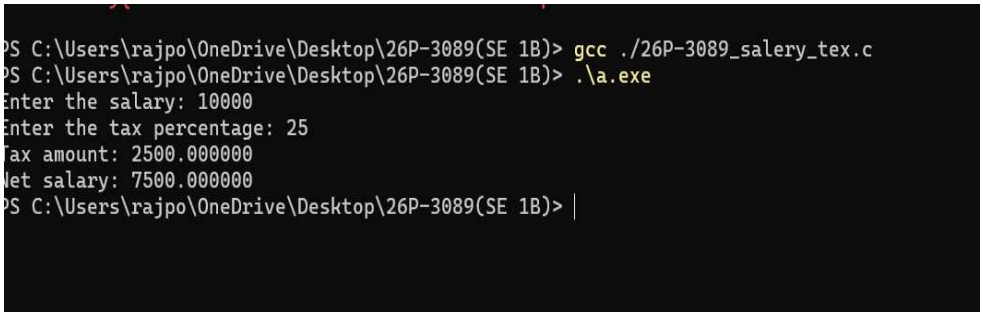
Code:-

```
  #include <stdio.h>
int main() {
 float salary, tax_rate, tax_amount, net_salary;
  printf("Enter the salary: ");
   scanf("%f", &salary);
  printf("Enter the tax percentage: ");
    scanf("%f", &tax_rate);
  tax_amount = (tax_rate / 100.0) * salary;
    net_salary = salary - tax_amount;
    printf("Tax amount: %f\n", tax_amount);
    printf("Net salary: %f\n", net_salary);
   return 0;}
```

NOTEPAD



OUTPUT ON TERMINAL



QUESTION#4 Converting Pennies into Coins

Code:-

```
#include <stdio.h>


int main() {
    int pennies = 368;
    int quarters, dimes;
    int nickels, remaining;


  quarters = pennies / 25;
    remaining = pennies % 25;
```

```
    dimes = remaining / 10;
    remaining = remaining % 10;

    nickels = remaining / 5;
    pennies = remaining % 5;

    printf("Quarters: %d\n", quarters);
    printf("Dimes: %d\n", dimes);
    printf("Nickels: %d\n", nickels);
    printf("Pennies: %d\n", pennies);

    return 0;
}
```

NOTEPAD

```c
#include <stdio.h>

int main() {
    int pennies = 368;
    int quarters, dimes;
    int nickels, remaining;

    quarters = pennies / 25;
    remaining = pennies % 25;

    dimes = remaining / 10;
    remaining = remaining % 10;

    nickels = remaining / 5;
    pennies = remaining % 5;

    printf("Quarters: %d\n", quarters);
    printf("Dimes: %d\n", dimes);
    printf("Nickels: %d\n", nickels);
    printf("Pennies: %d\n", pennies);

    return 0;
}
```

OUTPUT ON TERMINAL

```
PS C:\Users\rajpo\OneDrive\Desktop\26P-3089(SE 1B)> gcc .\26P_3089_pinnies_coins.c
PS C:\Users\rajpo\OneDrive\Desktop\26P-3089(SE 1B)> .\a.exe
Quarters: 14
Dimes: 1
Nickels: 1
Pennies: 3
PS C:\Users\rajpo\OneDrive\Desktop\26P-3089(SE 1B)>
```

QUESTION #6 Typecasting Before and After Division

Code:-

```c
#include <stdio.h>

int main() {

 int a;

  int b;

 printf("enter first intager ");

  scanf("%d",&a);

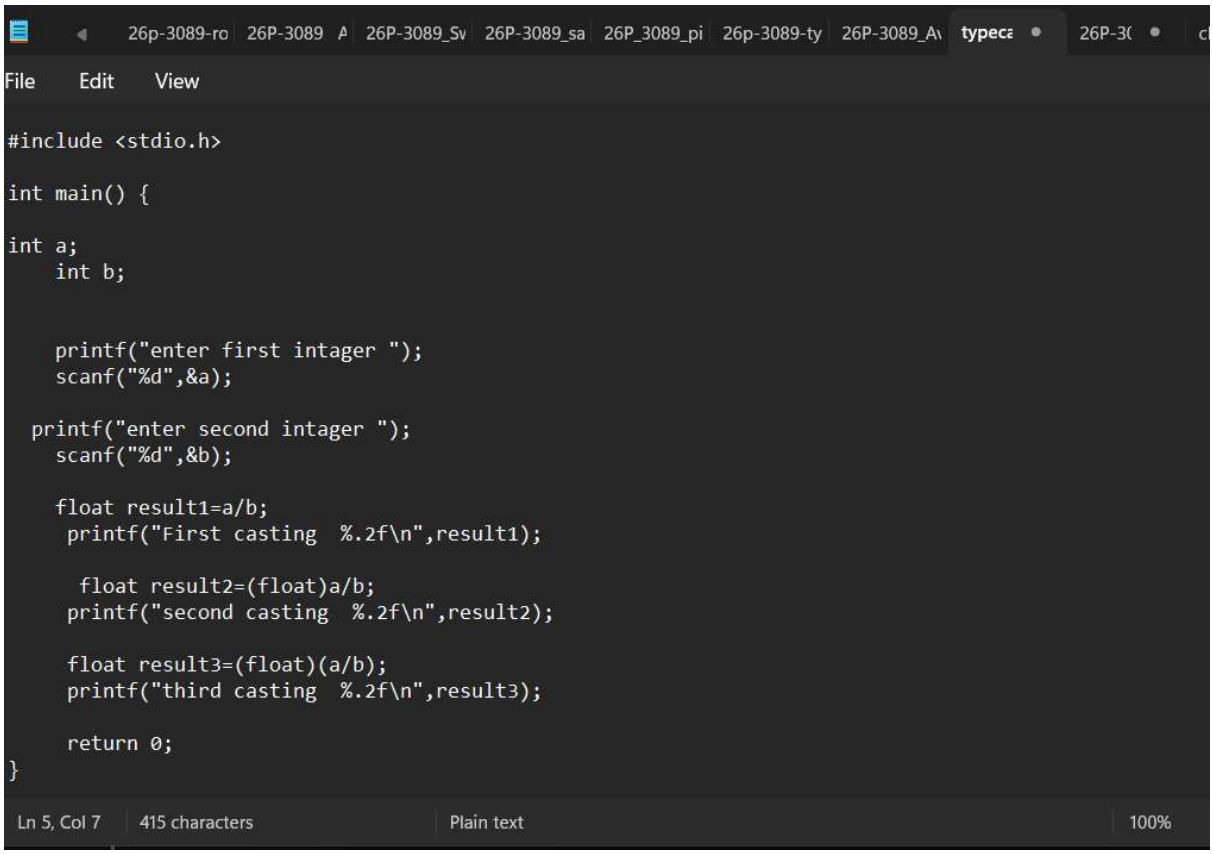
printf("enter second intager ");

  scanf("%d",&b);
```

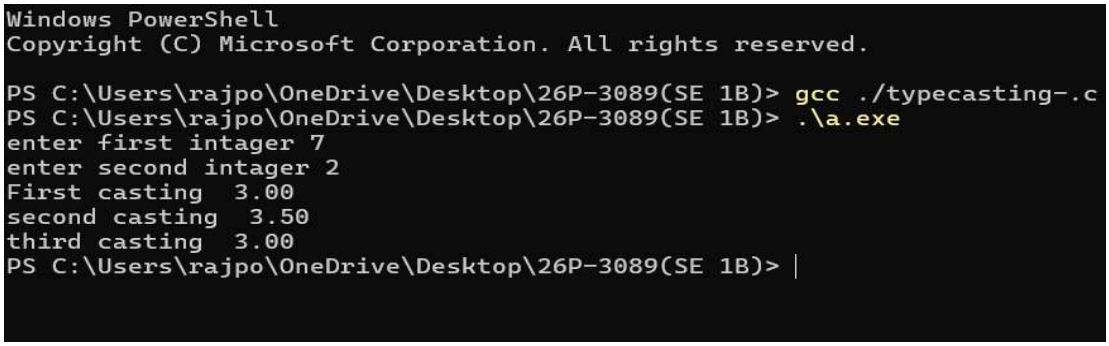
```
 float result1=a/b;
    printf("First casting  %.2f\n",result1);
  float result2=(float)a/b;
    printf("second casting  %.2f\n",result2);


    float result3=(float)(a/b);
    printf("third casting  %.2f\n",result3);
  return 0;
}
```

NOTEPAD



OUTPUT ON TERMINAL



QUESTION#7 Average Marks Using Typecasting

Code:-

```
#include <stdio.h>


int main() {
   int subject1;
  int subject2;
 int subject3;

   printf("Enter a first subject marks: ");
   scanf("%d", &subject1);

   printf("Enter a second subject marks: ");
   scanf("%d", &subject2);

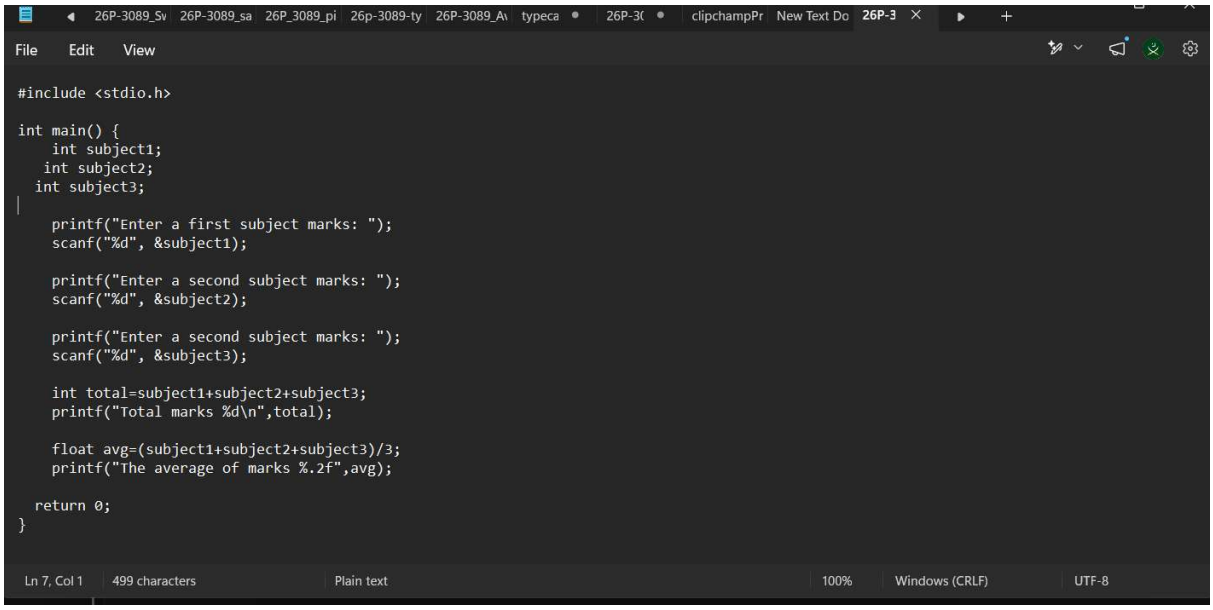
   printf("Enter a second subject marks: ");
   scanf("%d", &subject3);
```

```
    int total=subject1+subject2+subject3;
    printf("Total marks %d\n",total);

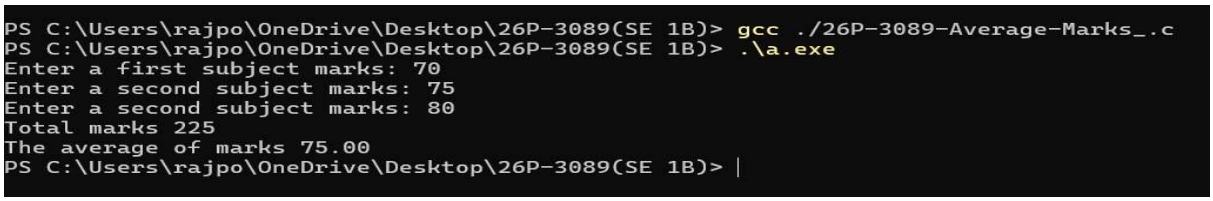
    float avg=(subject1+subject2+subject3)/3;
    printf("The average of marks %.2f",avg);

  return 0;
}
```

NOTEPAD



OUTPUT ON TERMINAL



QUESTION#8 Converting a Float to an Integer

Code:-

```
 #include <stdio.h>


int main() {

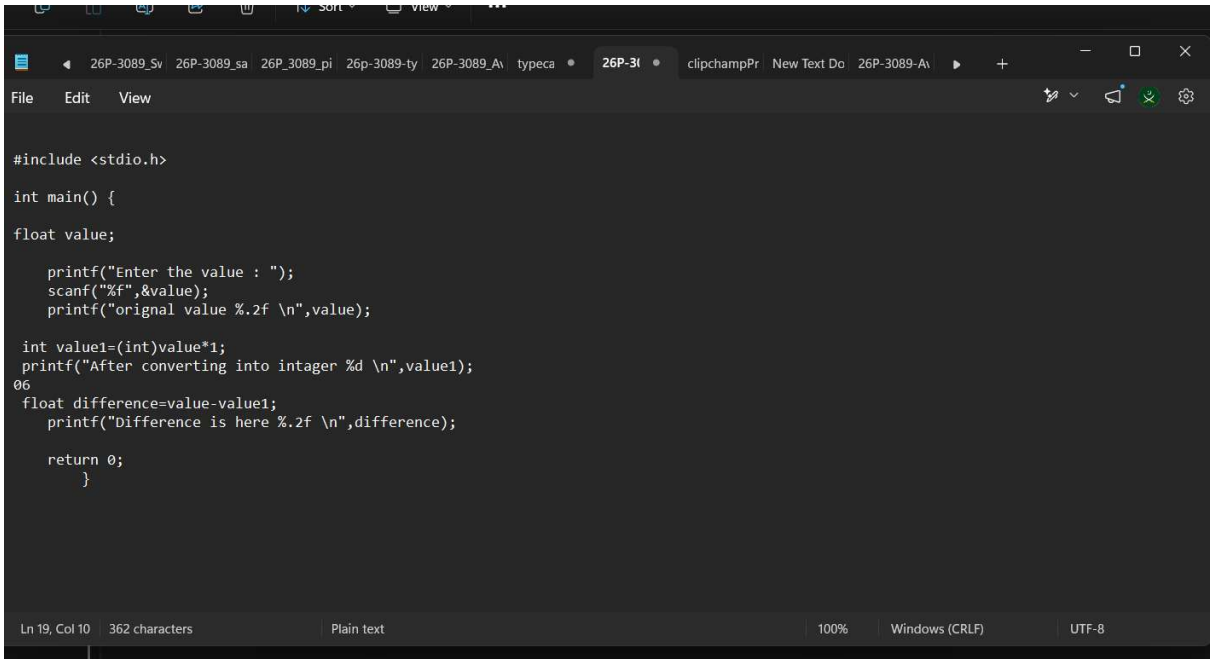

float value;


   printf("Enter the value : ");
   scanf("%f",&value);
   printf("orignal value %.2f \n",value);

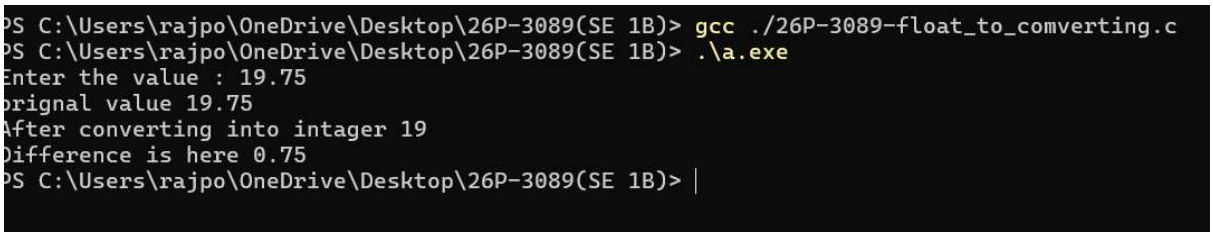

int value1=(int)value*1;
printf("After converting into intager %d \n",value1);
06
float difference=value-value1;
   printf("Difference is here %.2f \n",difference);


   return 0;
      }
```

NOTEPAD



OUTPUT ON TERMINAL



QUESTION#5 Roots of a Quadratic Equation

Code :-

#include <stdio.h>

int main() { int a, b, c; float disc; float x1, x2;

```
printf("Enter a value: ");
scanf("%d", &a);

printf("Enter b value: ");
scanf("%d", &b);

printf("Enter c value: ");
scanf("%d", &c);

disc = b * b - 4 * a * c;
```

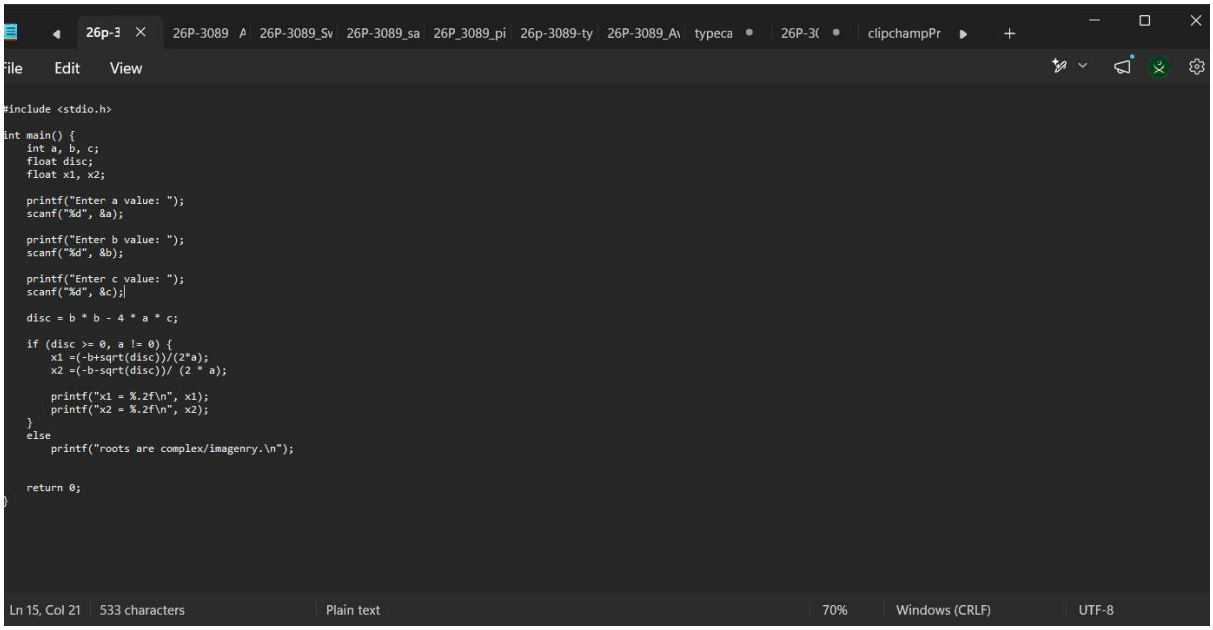
```
if (disc >= 0, a != 0) {
    x1 =(-b+sqrt(disc))/(2*a);
    x2 =(-b-sqrt(disc))/ (2 * a);

    printf("x1 = %.2f\n", x1);
    printf("x2 = %.2f\n", x2);
}
else
    printf("roots are complex/imagenry.\n");


return 0;


}
```

NOTEPAD



Output on terminal